

NextGen CRM Full Documentation and Upgrade Report

Daily startup guide, server map, database documentation, ERD/wireframe explanation, current feature report, missing features, and upgrade plan.

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Project folder: Z:\NextGen Projects\NextGen CRM

Main daily URL through Caddy: https://192.168.88.133:8443

Direct Laravel URL: http://127.0.0.1:8000

Database: MySQL on 127.0.0.1:3306, database name `nextgen_crm`

Update note, 21 May 2026: this CRM is a clone/rebuild of the current NextGen management system and is being upgraded into a cleaner Laravel + React CRM. It should preserve familiar operational screens while improving reliability, layout, filtering, renewal management, and service coverage.

1. Daily Start Guide in VS Code

Use this section every day when you forget how to start the CRM.

1. Open VS Code.
2. Open folder `Z:\NextGen Projects\NextGen CRM`.
3. Open the terminal with `Ctrl + ``.
4. If the terminal is not in the project folder, run:

```
cd "Z:\NextGen Projects\NextGen CRM"
```

Recommended daily development command:

```
composer run dev
```

This starts Laravel, the Laravel queue listener, and Vite frontend live reload.

Then open a second VS Code terminal and run Caddy:

```
caddy run --config Caddyfile
```

Open:

```
https://192.168.88.133:8443/login
```

The browser may show a certificate warning because this project uses Caddy internal TLS for local HTTPS. That is normal for local/internal development.

Alternative Manual Start

If `composer run dev` does not work, use three terminals.

```
php artisan serve --host=127.0.0.1 --port=8000
```

```
npm run dev -- --host 127.0.0.1
```

```
caddy run --config Caddyfile
```

2. Which Server You Are Using

Layer	What It Does	Current Setting
Laravel server	Runs PHP backend routes, controllers, auth, database access, and Inertia responses.	<code>127.0.0.1:8000</code>
Vite server	Runs React/Tailwind frontend development and live reload.	Usually <code>127.0.0.1:5173</code>
Caddy	Provides local HTTPS and reverse proxies browser traffic to Laravel.	<code>https://192.168.88.133:8443</code> to <code>127.0.0.1:8000</code>
MySQL	Stores CRM data.	<code>127.0.0.1:3306</code> , database <code>nextgen crm</code>
Laravel queue	Processes queued jobs.	Started by <code>composer run dev</code>

Current Caddyfile:

```
https://192.168.88.133:8443 {
  tls internal
  reverse_proxy 127.0.0.1:8000
}
```

3. Database You Are Using

The project uses MySQL. The current non-secret database settings are:

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE="nextgen crm"
DB_USERNAME=root
```

Laravel sessions, queues, and cache are also configured to use database tables:

```
SESSION_DRIVER=database
QUEUE_CONNECTION=database
CACHE_STORE=database
```

Useful database commands:

```
php artisan migrate
php artisan db:seed
php artisan migrate:fresh --seed
```

Only use `migrate:fresh --seed` on local development data because it deletes and recreates tables.

4. How the ERD Connects to the CRM

The ERD and wireframe files are:

```
docs\nextgen-crm-erd-wireframe.md
docs\nextgen-crm-erd-wireframe.svg
docs\nextgen-crm-erd-wireframe.png
```

The ERD is the map of the database. The wireframe is the map of the screens. Laravel migrations create the tables, Eloquent models connect PHP to the tables, controllers load the records, and React/Inertia pages render the screens.

```
customers
-> users
-> domain_hosting_requests
-> customer_subscriptions
    -> subscription_payments
    -> subscription_credits
```

Screen	Main Tables Used
Dashboard	customers, domain_hosting_requests, customer_subscriptions
Customers / Add Customer	customers
Domains	customers, domain_hosting_requests
Renewals	domain_hosting_requests, customer_subscriptions
Subscriptions	customer_subscriptions, subscription_payments, subscription_credits, customers
Support Requests	domain_hosting_requests, customers, users
Admin Settings	system_settings, users
Customer Portal	users, customers, domain_hosting_requests

5. Tools and Frameworks Used

Tool	Purpose
PHP 8.3	Backend programming language.
Laravel 13	Main backend framework.
Composer	PHP dependency manager and development script runner.
MySQL	Database server.
React 19	Frontend UI library.
Inertia.js	Connects Laravel controllers to React pages.
TypeScript	Safer frontend JavaScript.
Vite	Frontend development server and asset builder.
Tailwind CSS 4	Styling framework.
shadcn/Radix UI	UI component foundation.
Laravel Fortify	Login, password reset, email verification, and two-factor authentication.

Tool	Purpose
Caddy	Local HTTPS reverse proxy.
PHPUnit, ESLint, Prettier, Pint	Testing, linting, and formatting.
VS Code and Git	Editor and source control.

5A. NextGen Website Alignment

The public NextGen website at <https://nextgenpng.net/> positions Nextgen Technology as Papua New Guinea's premier domain hosting and ICT services provider. The CRM should track customers and subscriptions across the same service lines promoted on the website.

- Domain hosting and .pg domain registration.
- POP3 email accounts, website hosting, and email anti-spam protection.
- Internet Service Provider services.
- CCTV/security and surveillance.
- Document management solutions.
- Vehicle tracking systems.
- Audio visual systems.
- Network and infrastructure services.
- Website design and application development.

The CRM service categories already cover these areas through domain/hosting requests and subscription service types. The next upgrade should add clearer product/plan records for Basic, Standard, Value, and Premium hosting plans, including monthly prices and plan limits from the website.

6. Current Features Already Built

Latest UI and Behavior Fixes

- Create Subscription form fields were expanded and repaired so date, status, amount, service name, and reference inputs are no longer clipped.
- Subscription create form and subscription register now have working collapse/expand controls.
- Subscription register now supports horizontal scrolling when the table is wider than the available desktop viewport.
- Dashboard Total Suspended Customers now links to </customers?status=suspended>.
- Customer listing now supports a suspended-only server-side status filter.
- Renewals page now has summary cards, visible-row select/clear actions, clearer month search, and selected-count behavior.

Authentication and Roles

- Login, registration routes, password reset, email verification, two-factor support.
- Admin, staff, and customer roles.
- Customer users can link to customer records.

Dashboard

- Customer, domain, suspended customer, support request, and domain registration metrics.
- Recent request list.
- Staff dashboard and customer portal behavior.

Customer Management

- List, create, update, and admin delete.
- Company, contact, email, phone, industry, status, website, address, notes, and follow-up date.

Requests and Domains

- Hosting/service request index.
- Domain listing, domain registration listing, support request listing, and detail pages.

	• Assignee, quote, renewal date, requirements, and internal notes.
Subscriptions • Service type, service name, reference, status, start date, expiry date, renewal cycle, amount, and notes. • Linked customer and history records.	Payments and Credits • Payment records, invoice numbers, payment references, and file uploads. • Service credits calculate months and extend expiry dates.

7. Important CRM Routes

URL	Purpose
/login	Login
/dashboard	Dashboard or customer portal behavior
/customers	Customer list
/add-customer	Add customer
/domains	Domain list
/renewals	Renewal monitoring
/subscriptions	Subscription management
/hosting-requests	Service request management
/support-requests	Support request list
/domain-registrations	Domain registration workflow
/bulk-email-validator	Bulk email tool
/admin-settings	Branding and email settings
/users	Admin user management

8. Demo Accounts

admin@nextgenpng.net
staff@nextgenpng.net
customer@nextgenpng.net

Password for seeded demo accounts:
password

Change seeded passwords before production use.

9. What Is Missing or Needs Improvement

High Priority

1. Automated renewal reminders by email, SMS, or dashboard notification.
2. Production backup process for MySQL and uploaded invoice files.
3. Production deployment plan with domain name, public SSL, queue worker, scheduler, and database backup.
4. Audit log for deletes, subscription changes, credit changes, and SMTP settings updates.
5. Stronger staff/admin/customer permission policy.
6. Customer portal invoice/payment visibility decision.
7. Password policy and forced password change for seeded/demo users.
8. Product/plan table for NextGen website hosting plans: Basic, Standard, Value, Premium.
9. Direct migration/import plan from the current cloned management system into the upgraded CRM schema.

Business Features to Add

1. Invoice generator and printable receipt.
2. Payment status per subscription: paid, partial, overdue, unpaid.
3. Customer statement report.
4. Expiring soon report.
5. Overdue invoice report.
6. Service credit approval workflow.
7. Ticket comments and activity timeline.
8. File attachments for support requests.
9. Task assignment and due-date reminders for staff.
10. Customer communication history.
11. Quote/proposal module.
12. Product/service catalog with pricing.
13. Dashboard charts for revenue, renewals, support workload, and service categories.
14. Hosting plan catalog matching NextGen public website prices and limits.
15. Domain availability/registration workflow integration if NextGen wants end-to-end .pg registration tracking.
16. Email settings lookup/help page for customer support staff.

Technical Improvements

1. Add automated browser tests for critical workflows.
2. Add API/resource tests for permissions.
3. Add indexes after production data grows.
4. Add soft deletes for business records.
5. Add file storage policy for uploads.
6. Add scheduled command for renewal reminders.
7. Add queue worker service for production.
8. Review `.env.example` to match Caddy and MySQL setup.
9. Add a one-command Windows startup script.
10. Add deployment documentation for a real server.

10. Upgrade Plan

Phase	Work
Phase 1: Stabilize daily development	Create one reliable startup script, align README/start files/Caddyfile, choose one port, document daily startup, and keep cloned legacy navigation familiar while progressively replacing broken

Phase	Work
	layouts.
Phase 2: Protect data	Add MySQL backup, uploaded file backup, restore instructions, soft deletes, and audit log.
Phase 3: Improve CRM operations	Add renewal reminders, payment status, statements, PDF invoices, request attachments, staff tasks, dashboard alerts, and a service plan catalog aligned to nextgenpng.net.
Phase 4: Production readiness	Use a real domain, public SSL, queue worker service, scheduler, production env, real SMTP, strong passwords, and monitoring.

11. Quality Commands

```
npm run types:check
npm run lint:check
npm run format:check
npm run build
php artisan test
composer run lint
composer run lint:check
php artisan optimize:clear
php artisan route:list
```

12. Troubleshooting

Page Does Not Open

```
php artisan serve --host=127.0.0.1 --port=8000
caddy run --config Caddyfile
```

Frontend Changes Do Not Show

```
npm run dev
npm run build
php artisan optimize:clear
```

Database Error

```
php artisan migrate
```

Login Does Not Work

```
php artisan db:seed
```

13. Daily Programmer Checklist

1. Open VS Code.
2. Open Z:\NextGen Projects\NextGen CRM.
3. Open terminal.
4. Run: composer run dev
5. Open second terminal.
6. Run: caddy run --config Caddyfile
7. Open: <https://192.168.88.133:8443/login>
8. Login with admin account.
9. Before coding run: git status --short

```
10. After coding run tests/build.  
11. Write what changed.  
12. Commit when ready.
```

14. Expert Notes

- Keep one source of truth for ports. The current Caddyfile expects Laravel on port 8000. Some older notes mention 8001.
- `composer run dev` is the best development start command because it starts Laravel, queue, and Vite together.
- Caddy should stay in a separate terminal because it is the reverse proxy.
- Do not expose `.env` passwords in Git or documentation.
- Do not use `php artisan migrate:fresh --seed` unless you are comfortable deleting local data.
- Before production launch, add backups, audit logs, real SMTP, strong passwords, scheduler, queue service, and HTTPS with a real domain.